

Foundation GCSE: Decimal Arithmetic

Name: _____

Class: _____

Instructions: Answer all questions. Questions are ordered from easiest (1) to hardest (20). Working out should be shown where appropriate.

1. Calculate: $3.4 + 2.1$
2. Calculate: $5.8 - 2.3$
3. Calculate: $7.2 + 1.65$
4. Calculate: $6.4 - 2.8$
5. Calculate: 3.2×4
6. Calculate: $15.6 \div 3$
7. Calculate: 0.4×0.5
8. Calculate: $4.8 \div 0.2$
9. Work out the value of 1.2^2
10. Find the square root of 0.09
11. Insert the correct symbol ($<$, $>$, or $=$) between: $0.45 \dots 0.405$
12. Round 47.852 to 1 decimal place
13. Round 0.0384 to 2 significant figures
14. Calculate: $2.5 + 3.1 \times 2$
15. Calculate: $(4.2 - 1.2) \div 0.5$
16. Estimate the value of $\frac{21.4 \times 8.9}{3.2}$ by rounding each number to 1 significant figure
17. Express 0.35 as a fraction in its simplest form
18. Convert the recurring decimal $0.\dot{3}\dot{6}$ into a fraction in its simplest form
19. Petrol costs £1.42 per litre. A driver fills up 45.5 litres and pays with a £100 note. How much change do they receive?
20. Calculate the exact value of $\frac{3.6 \times 10^{-2} + 0.0024}{0.001}$

Answers & Worked Solutions

1. 5.5
Align decimals: $3.4 + 2.1 = 5.5$.
2. 3.5
Subtract tenths and units: $5.8 - 2.3 = 3.5$.
3. 8.85
Pad with zero: $7.20 + 1.65 = 8.85$.
4. 3.6
Borrow 1: $6.4 - 2.8 = 3.6$.
5. 12.8
Multiply ignoring decimal: $32 \times 4 = 128$, then adjust to 12.8.
6. 5.2
Divide normally: $15.6 \div 3 = 5.2$.
7. 0.2
 $4 \times 5 = 20$, with two decimal places gives 0.20 or 0.2.
8. 24
Multiply both by 10: $48 \div 2 = 24$.
9. 1.44
 $12 \times 12 = 144$, two decimal places gives 1.44.
10. 0.3
Since $0.3 \times 0.3 = 0.09$, the root is 0.3.
11. $>$
Compare digit by digit: 0.45 is greater than 0.405.
12. 47.9
Look at the second decimal digit (5), round up.
13. 0.038
First two non-zero digits are 3 and 8, followed by 4 (round down).
14. 8.7
BODMAS: Multiply first ($3.1 \times 2 = 6.2$), then add 2.5.
15. 6
Brackets first (3.0), then $3.0 \div 0.5 = 6$.
16. 60
Round to 1 sf: $\frac{20 \times 9}{3} = \frac{180}{3} = 60$.
17. $\frac{7}{20}$
Fraction is $\frac{35}{100}$, divide top and bottom by 5.
18. $\frac{4}{11}$
Let $x = 0.3636\dots$, $100x = 36.3636\dots$, $99x = 36$, simplify $\frac{36}{99} = \frac{4}{11}$.
19. £35.39
Cost = $45.5 \times 1.42 = £64.61$. Change = $100 - 64.61 = £35.39$.
20. 38.4
Numerator: $0.036 + 0.0024 = 0.0384$. Divide by 0.001 shifts decimal 3 places right: 38.4.